MCS 481 Project Three : Voronoi Diagrams of IL Tornadoes
due Friday 6 November 2026 at noon

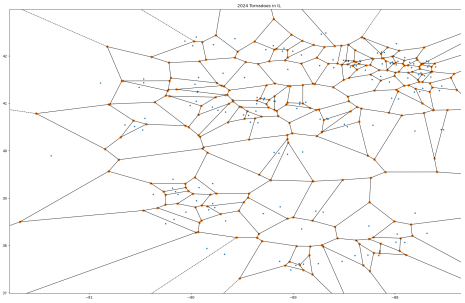
The goal of this project is to study Voronoi diagrams, using historical data on tornadoes in Illinois.

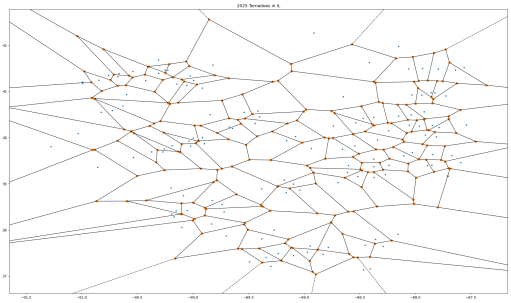
1. Tornadoes in Illinois

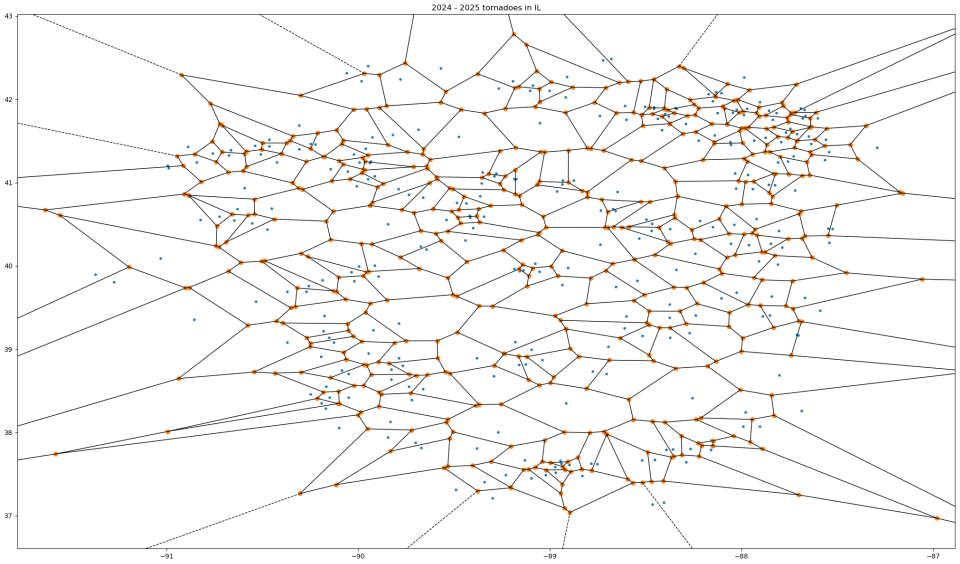
The Storm Prediction Center (SPC) of the the National Oceanic and Atmospheric Administration (NOAA) provides comma separated value (csv) files with historical data of tornadoes, available at

`https://www.spc.noaa.gov/wcm/#data`

in files `2024_torn.csv` and `2025_torn.csv`. Using longitude and latitude as the (x, y)-coordinates on file, Voronoi diagrams, made with `matplotlib`, and the `spatial` module of `scipy` are shown below.







Assignment One. Download the coordinates of tornadoes in Illinois and make the above plots.

1. Compare the time it takes to construct Voronoi diagrams with the time it takes to load and construct the input for the construction of the Voronoi diagrams.
2. Compare the construction times for each diagram. The 2024-2025 Voronoi diagram has about twice the amount of input as the other two diagrams. Does it take twice as long to make?

2. Making Queries

With Voronoi diagrams we can study the geometric shape of where tornadoes happened.

Assignment Two. Use the Voronoi diagrams to make queries.

1. Given a position, specified by longitude and latitude, with the Voronoi diagram we can locate the site nearest to the given position where a tornado occurred.

   How sensitive is the outcome of such queries to the closeness of the sites?
   Answer this question with illustrations of appropriate queries on the data.

2. What is the closest cluster of tornado locations?
   Where are the distances between the sites the largest?

   (a) Develop an algorithm to answer these questions and demonstrate its correctness on the data.

   (b) Express the cost of your algorithm in terms of the number of sites.

3. The Incremental Construction of Voronoi Diagrams

Given the Voronoi diagram for the tornadoes in 2024, we consider the incremental addition of the locations of the tornadoes in 2025. Adding sites one of the other might lead to the discovery of a pattern where and when tornadoes are likely to happen.

Assignment Three. Add the data points of the 2025 tornadoes to the Voronoi diagram of the 2024 tornadoes.

1. What is the cost of this incremental construction?

2. Define an algorithm to add one new site to a Voronoi diagram.

3. Express the cost of adding one site to a Voronoi diagram as a function of the number of sites.

4. the Deadline is Friday 6 November 2026 at noon

Upload your answers to gradescope before noon on Friday 6 November 2026.
Your solution consists of

1. The specification of your computers and the version of the software used.

2. The code for the programs you used in the experiments.

3. All relevant plots, preferably embedded in the output cells of the submitted Jupyter notebook, or otherwise uploaded as separate png files.

4. Tables with properly formatting of the essential characteristics for each experiment: the dimension of input and output, and the running time. Raw data may be provided as an appendix.

5. Your interpretation of the results and conclusions.

You may work in pairs for this project.

If you work in a pair, then one member of the pair must email the name of the partner to `janv@uic.edu` before 5pm on Friday 31 October.

If you have questions or difficulties with the project, feel free to ask questions during office hour.

As an alternative to the final exam, there is the final research project.

MCS 481 Resarch Project : a topic in computational geometry
as a replacement of the final exam

The topic in the research project is open to your own interest and could vary between a programming project and a theoretical study. In particular, you could select to do one of the following:

1. Write your own code to explore an algorithm we covered in class.
2. Use CGAL to explore algorithms not covered in the projects.
3. Use other software, e.g.: SageMath, Qhull, for computations.
4. Select a theoretical topic, from a section we did not cover in the textbook.
5. Study a paper from the computational geometry literature.

You must announce to me in an email to `janv@uic.edu` the selection of your topic by 5pm on Monday 10 November. Write a paragraph or two stating the goals of the project.

Deadlines

The replacement of the final exam by a final research project has several milestones, outlined in the deadlines below:

1. Monday 9 November: commitment to a topic
2. Monday 23 November: first draft of a technical report
3. Monday 30 November: ready for an in-class presentation
4. Tuesday 8 December: final technical report

The day of the final exam is the deadline for the final report.

This project must be an individual effort. Collaborations are not allowed.

If you have questions or difficulties with the project, feel free to come to my office for help.